SkyLink International Airport

Airline Delay & Resource Optimization

Business Analyst Portfolio Project 1

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EXCEL | BUSINESS ANALYSIS | DASHBOARD | EXECUTIVE SUMMARY

Business Problem

Why was this project needed?

Flight delays impacting punctuality

Uneven gate & stand utilization

Passenger flow congestion

Resource allocation inefficiencies

Business Need: Identify operational bottlenecks and recommend improvements using historical airport data.

Objectives

- Identify major delay bottlenecks
- Evaluate gate & stand utilization
- Analyze passenger flow
- Detect peak congestion hours
- Recommend operational improvements

Project Overview

Dataset- Third party Operational dataset

5,921
Records

124
Columns

86
Cleaned Columns

Tool Used

Microsoft Excel | Pivot Tables| Charts

KPI

11 KPIs Developed

Data Quality Findings

📌 **Impact :**

Passenger count: Exceeded aircraft capacity

90–95% counter allocation missing, counter utilization unreliable

Tow-date mismatch, affecting date analysis

Key Business Requirements

Operations Manager

➡️ Identify flight delay patterns

Ground Operations

➡️ Monitor gate & stand utilization

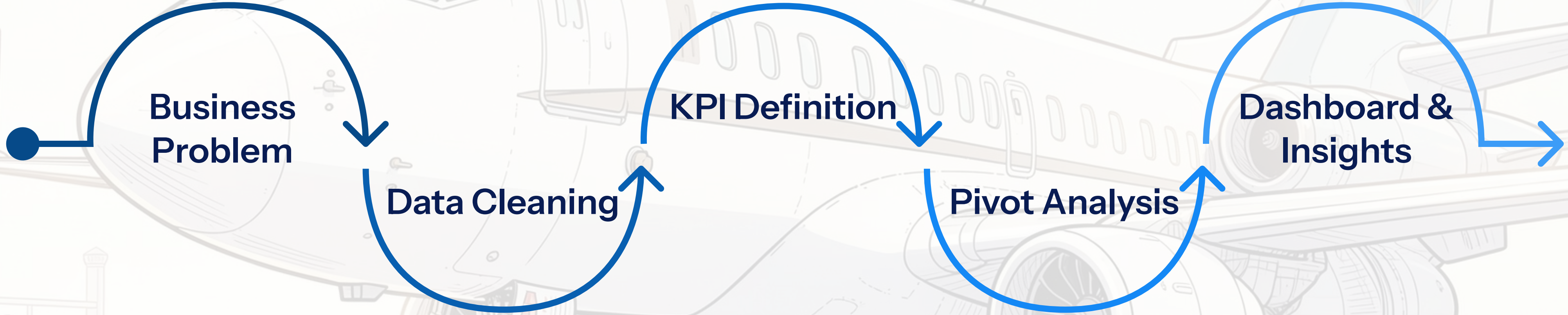
Resource Planner

➡️ Analyze passenger demand

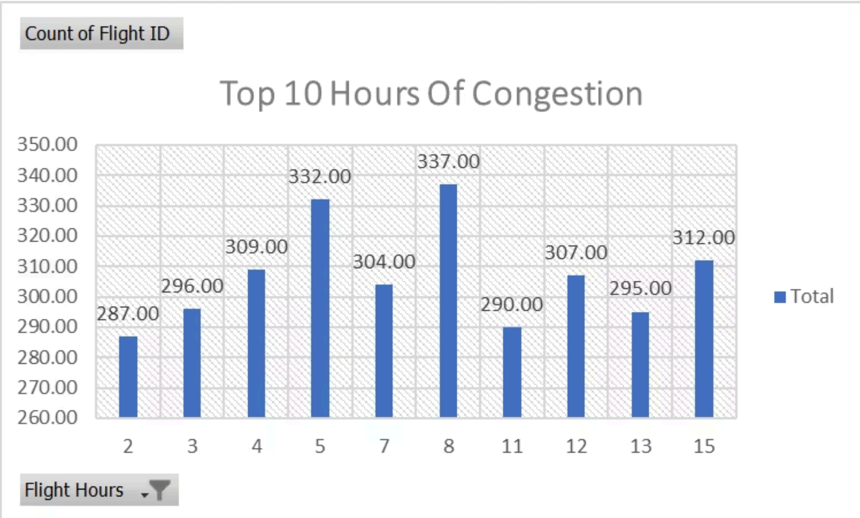
Airport Leadership

➡️ Improve operational efficiency

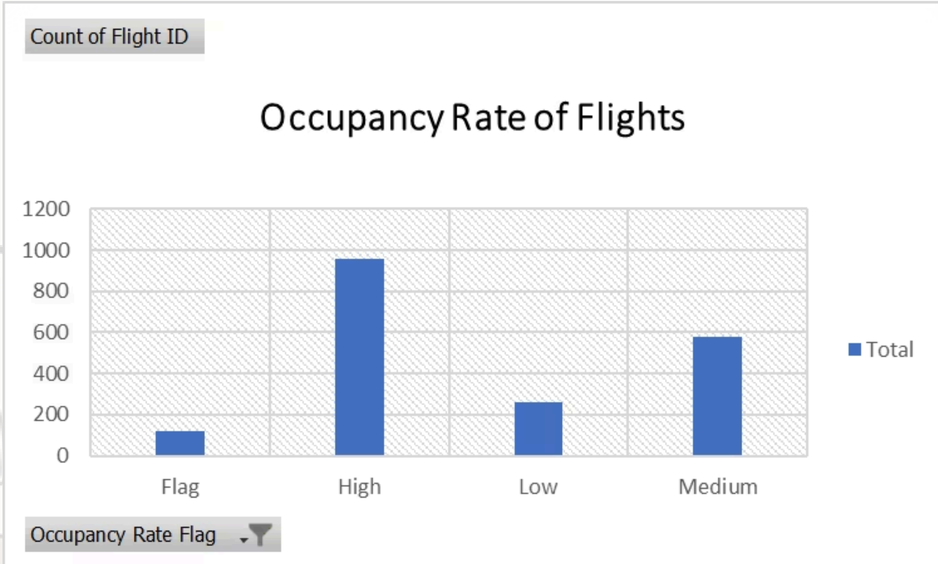
Analysis Process



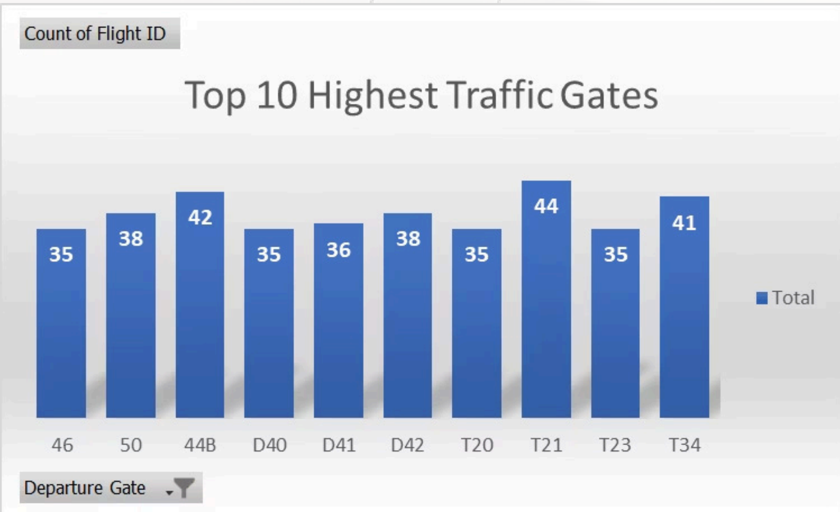
Dashboard



Count of Flight ID
Top 10 Hours Of Congestion
350.00
340.00
330.00
320.00
310.00
300.00
290.00
280.00
270.00
260.00
287.00
296.00
309.00
332.00
304.00
337.00
290.00
307.00
295.00
312.00
2
3
4
5
7
8
11
12
13
15
Total
Flight Hours



Count of Flight ID
Occupancy Rate of Flights
1200
1000
800
600
400
200
0
Flag
High
Low
Medium
Total
Occupancy Rate Flag

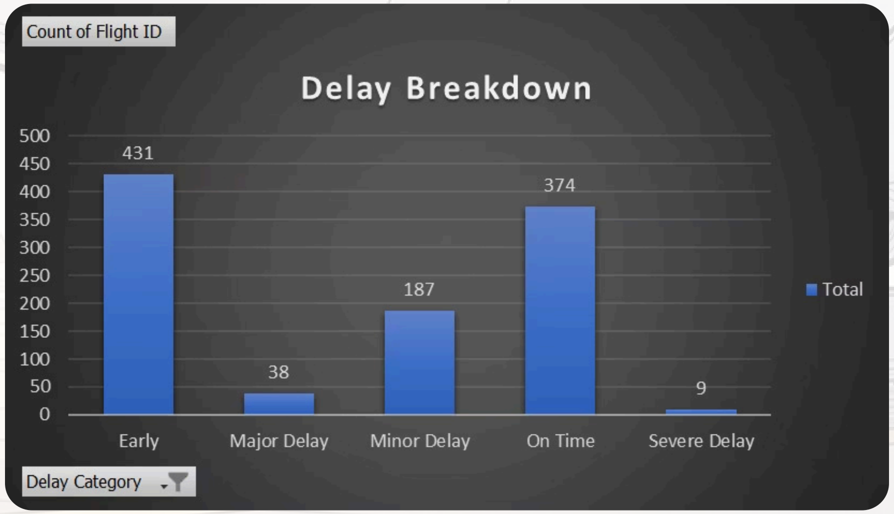


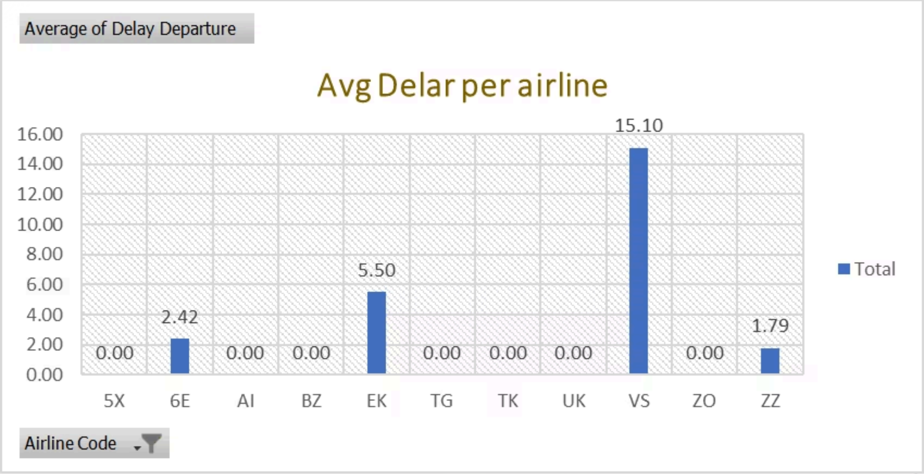
Count of Flight ID
Top 10 Highest Traffic Gates
35
38
42
35
36
38
35
44
35
41
46
50
44B
D40
D41
D42
T20
T21
T23
T34
Total
Departure Gate

Count of Flight ID
Top 7 Congested Parking Stands
80
70
60
50
40
30
20
10
0
A06
C29L
C29R
202
D50
142
141
Total
Parking Stand

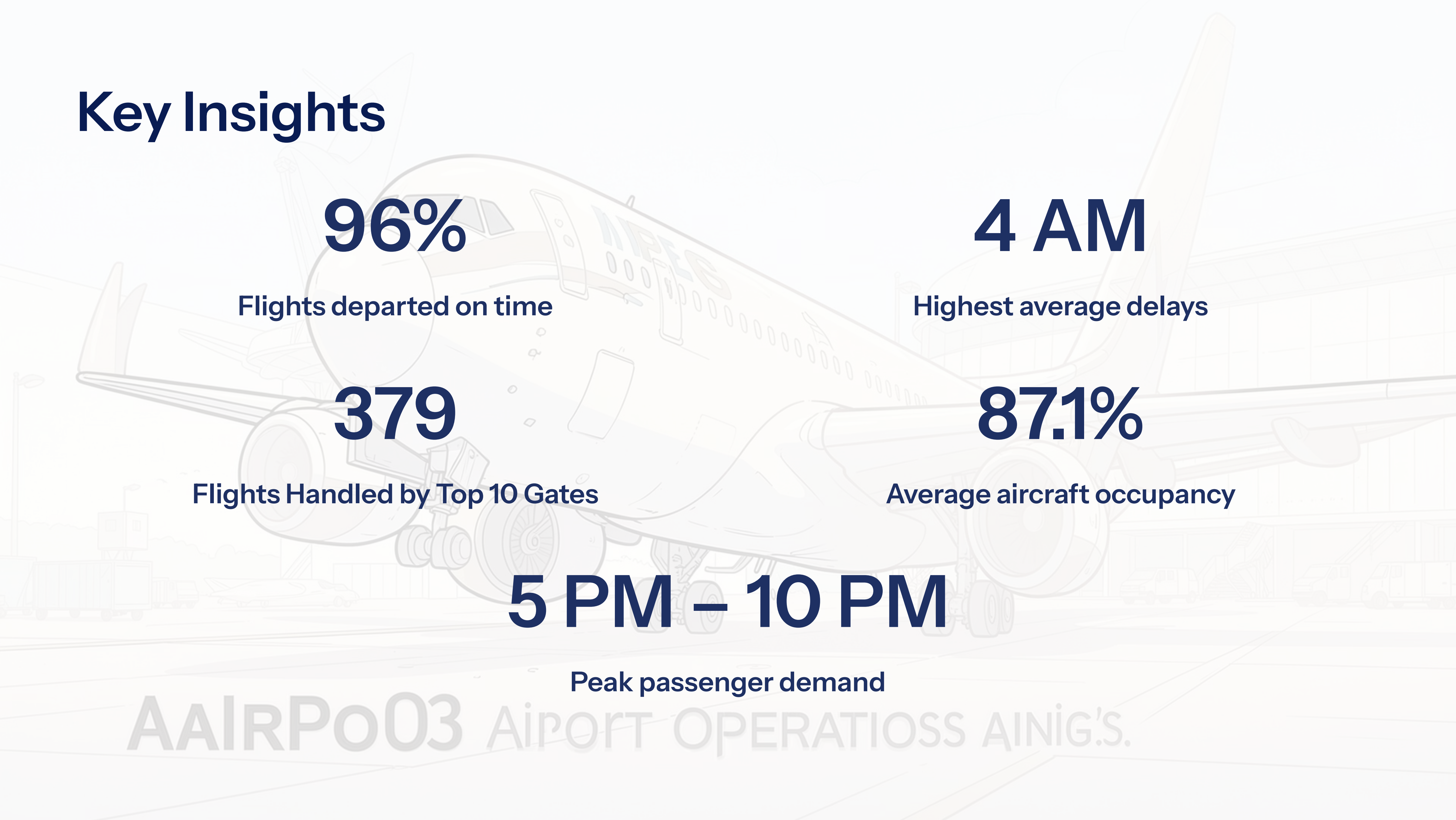
Gate Congestion Heatmap

Count of Flight ID	Hour									
Gate	**2**	**3**	**4**	**5**	**7**	**8**	**11**	**12**	**13**	**15**
46		2.00	2.00	1.00	5.00	2.00	2.00	1.00	3.00	4.00
48	2.00		1.00	1.00		3.00	3.00	3.00	5.00	2.00
44B	1.00	4.00		2.00	4.00	1.00	2.00	3.00	5.00	5.00
D34	2.00	6.00	1.00	2.00	3.00	2.00		1.00	1.00	2.00
D35	4.00	2.00	1.00	2.00	1.00	4.00	2.00	2.00		3.00
D40	2.00	6.00	2.00	1.00		2.00		4.00	3.00	
D42		4.00	4.00	2.00	2.00	3.00	1.00	1.00	4.00	1.00
T21	3.00		5.00	1.00	2.00	1.00		2.00	4.00	3.00
T23	2.00	1.00	2.00	2.00	1.00	4.00	4.00		2.00	2.00
T34	3.00	2.00	2.00	2.00	4.00	3.00	2.00		2.00	





Key Insights



96%

Flights departed on time

4 AM

Highest average delays

379

Flights Handled by Top 10 Gates

87.1%

Average aircraft occupancy

5 PM – 10 PM

Peak passenger demand

Recommendations

Optimize the 4 AM turnaround process

Redistribute flights from high-traffic gates during peak hours.

Review performance of high-delay airlines (VS, EK, 6E, ZZ).

Streamline international departure process

Increase staffing during the 5–10 PM passenger peak.

Implement stronger data validation controls for reliable reporting.

Expected Business Impact

Reduced flight delays

Faster turnaround

Better gate utilization

Improved passenger experience

Better operational planning

More efficient resource allocation

Business Analyst Skills

Business Analysis

- Requirement Gathering
- Stakeholder Analysis
- KPI Definition
- Business Recommendations

Data Analysis

- Data Cleaning
- Excel
- Pivot Tables
- Dashboard Development
- Root Cause Analysis

Deliverables

- 📄 Business Requirements Document
- 📄 Executive Summary
- 📊 Excel Dashboard
- 📈 Business Presentation